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June 12, 1975
SUPERSEDING
Int. Fed. Spec. O-C-00940A(GSA-FSS)
December 31, 1969

FEDERAL SPECIFICATION

CYCLOHEXYLAMINE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of cyclohexylamine (see 6.1).

1.2 Classification. Cyclohexylamine shall be of the following types, as specified (see 6.2):

Type I - 98.0 percent solution.
Type II - 60.0 percent solution.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-D-705 - Drum: Metal Shipping, Steel, (Over 12 and under 55 Gallon).
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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Federal Regulation:

49 CFR 173 Subpart H - Marking and Labeling of Hazardous Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Cyclohexylamine shall be a liquid.

3.2 Purity. Cyclohexylamine shall be of the purity in table I when tested as specified in 4.3.1.

TABLE I. Purity requirement for cyclohexylamine

Requirement	Types I	II
Purity, weight percent, minimum	98.0	60.0

3.3 Workmanship. Cyclohexylamine shall be clear, visibly free from foreign matter and suitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery.

4.2.1.1 Lot. For this examination, a lot shall be cyclohexylamine, fully packed, marked, and offered for examination at one time. Lot size shall be expressed in number of filled unit containers.

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4.2.1.2 Sampling plan. Sample unit for this examination shall be one unit container. Sampling shall be conducted in accordance with MIL-STD-105. Sample size, acceptance and rejection numbers shall be determined by general inspection level II and acceptable quality level (AQL) 6.5 percent defective, respectively. Defects shall be scored in accordance with table II.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; not as specified.
Packing	Not as specified.
Contents	Net contents less than specified.
Workmanship	Split, crack, hole; closure incomplete or inadequate; leakage or seepage from any part of container.

4.2.2 Examination of material and workmanship.

4.2.2.1 Lot. For this examination, a lot shall be cyclohexylamine produced by one manufacturer, at one plant, from the same raw materials, within a 24-hour period, by the same manufacturing process, and offered for examination at one time. Lot size shall be expressed in number of pounds of cyclohexylamine.

4.2.2.2 Sampling plan. Sample size for this examination shall be two pounds (approximately one liter). Sample shall be selected by drawing equal portions at random throughout the lot. Defects shall be determined in accordance with table III. Lot shall be rejected if sample contains one defect.

TABLE III. Examination of material and workmanship

Examine	Defect
Material	Not a liquid.
Workmanship	Not clear; foreign matter visibly present.

4.2.3 Testing.

4.2.3.1 Lot. A lot shall be as specified in 4.2.2.1.

4.2.3.2 Sampling plan. Sample size for testing shall be two pounds, selected as specified in 4.2.2.2. Sample portions shall be thoroughly mixed to yield a homogeneous composite and stored in a clean, dry, glass container with identifying lot number and date. All testing shall be performed on the composite sample. Lot shall be rejected if the composite fails to meet any specified requirement.

4.3 Test method.

4.3.1 Purity. Into a one-liter volumetric flask, place approximately 10g of cyclohexylamine, weighed to the nearest 0.01g. Dilute with distilled water and mix thoroughly until homogeneous. Transfer a 25-ml aliquot to a 250-ml wide-mouth erlenmeyer flask and titrate with standardized 0.1N-hydrochloric acid against methyl red indicator. Calculate purity of cyclohexylamine by:

$$\% \text{ cyclohexylamine} = \frac{N \times V \times 397}{W}, \text{ where}$$

N = normality of hydrochloric acid.
V = volume, ml, of hydrochloric acid.
W = weight, g, of sample.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B.

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5.1.1.1 Type I.

5.1.1.1.1 One hundred fifteen pound quantities. One hundred fifteen (115) pounds of type I shall be furnished in a 16-gallon steel shipping drum conforming to PPP-D-705, type II, except that the exterior surfaces may have a commercial coating.

5.1.1.1.2 Three hundred seventy-five pound quantities. Three hundred seventy-five (375) pounds of type I shall be furnished in a 55-gallon steel drum conforming to PPP-D-729, type II, except that the exterior surfaces may have a commercial coating.

5.1.1.2 Type II.

5.1.1.2.1 One hundred twenty pound quantities. One hundred twenty (120) pounds of type II shall be furnished in a 16-gallon steel shipping drum conforming to PPP-D-705, type II, except that the exterior surfaces may have a commercial coating.

5.1.1.2.2 Four hundred pound quantities. Four hundred (400) pounds of type II shall be furnished in a 55-gallon steel drum conforming to PPP-D-729, type II, except that the exterior surfaces may have a commercial coating.

5.1.2 Level C. Cyclohexylamine, in quantities as specified, shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall be in compliance with Uniform Freight Classification, or National Motor Freight Classification rules, as applicable.

5.2 Marking. In addition to any special marking required by the procurement document, marking shall be in accordance with Fed. Std. No. 123, or MIL-STD-129, as specified (see 6.2).

5.2.1 Labeling. In addition to any special labeling required by the procurement document, labeling shall be in accordance with 49 CFR 173 subpart H.

6. NOTES

6.1 Intended use. Cyclohexylamine is intended for use as a corrosion inhibitor in low pressure heating plants. It volatilizes with steam, circulates through the lines, and returns with the condensate to the boiler. Cyclohexylamine neutralizes acidity resulting from dissolved carbon dioxide and maintains alkalinity to inhibit corrosion in boilers, pipes, and radiators.

6.1.1 Types. Selection of type may depend upon environmental conditions of storage and handling. To aid selection of type, physical properties are provided in table IV below.

TABLE IV. Physical properties of cyclohexylamine

Property	I	II
Flash point, TOC, C°	32	57
Freezing point, C°	-18	-1
Specific gravity @ 20°/20° C.	0.869	0.948

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable level, types, and size of packing (see 5.1).
(c) Marking and labeling documents required (see 5.2 and 5.2.1).

Military coordinating activity:

Army - GS

Preparing activity:

GSA-FSS

Civil agency coordinating activities:

GSA-FSS
INT-MIN
VA-DMS

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